VIRAJ CHOUDHARY

BTech in Artificial Intelligence | 2023 - 2027
github.com/virajchoudhary

Education

Mukesh Patel School of Technology Management & Engineering, NMIMS 2023 - 2027
Bachelor of Technology in Artificial Intelligence Semester 6 CGPA: 3.25/4.00

Billabong High International School 2021 - 2023
CBSE Senior School Certificate (Class XII): 67.60

Billabong High International School 2021
CBSE Secondary School Certificate (Class X): 81.80

Projects

AtlasOps - Multi-Agent SRE Research Platform 2026
- Extended an inherited four-agent SRE platform with controls for proposed Kubernetes actions and history APIs.
- Built authenticated, digest-bound FastAPI approvals; added spoofed-identity and audit-integrity tests.

Kesten - Stock Prediction & Portfolio Analytics 2026
- Contributed PyTorch LSTMs, ARIMA/GARCH baselines and historical backtests to Django REST/Streamlit.
- Built Next.js UI and authenticated Django APIs with owner-scoped watchlists; added API/browser tests.
- Kesten is a working name for a local-only continuation; legal and domain clearance pending.

GridShift - Grid Forecasting and GPU Scheduling Simulation 2026
- Team study: SARIMA cut mean MAE 16.03% versus persistence across five horizons on the frozen 2025 test; 70,128 normalized GB half-hour intervals, 69,950 observed.
- Team simulation: Calendar-mean versus earliest scheduling cut modeled CO2 12.41% for 2,913 GPU jobs; zero deadline misses.

Internships

Kotak Mutual Fund | Software Engineering Intern May 2026 - Present
- Developed Python/FastAPI and React dashboards for internal sales, financial analytics and MIS reporting.
- Built Excel/CSV ingestion and validation pipelines, REST APIs and automated reporting exports.
- Added regression tests and security controls; worked with business and IT on deployment readiness.

Certifications & Publications

LEARN-Net - Explainable Diabetic Retinopathy Grading 2025 - Present
Co-author | SN Computer Science manuscript under revision
- Co-authored four-class APTOS 2019 research combining LAB-CLAHE preprocessing, residual CNN blocks, CBAM attention and Grad-CAM explanations.
- Team manuscript reports a 2.26M-parameter model with QWK 0.9097 and 82.51% accuracy; no external clinical validation.
- Added FastAPI health, prediction and invalid-upload tests; integrated React prediction and Grad-CAM requests through API proxy routes.

Double-Heston Bitcoin Option Pricing 2026
Co-author | Current research manuscript
- Built a Fourier pricer and independent benchmark; 36/36 controlled cases passed in historical validation.
- Team PINN study: 1,102 authentic Deribit trades, ten retrospective episodes; $0.34-$0.59 RMSE vs Fourier.
- Team results at fixed 45%/60% initial volatility: beat Single Heston in 10/10 episodes; 0/10 at 28%, reverse Black-Scholes ranking. No market calibration or trading signal.

AWS Academy Graduate - Cloud Architecting April 2026
AWS Academy Graduate - Cloud Foundations March 2026

CORE SKILLS

Technical Skills: Python, C++, JavaScript, SQL; PyTorch, scikit-learn, Pandas, NumPy, SciPy; computer vision and time-series forecasting

Development: FastAPI, React, Django REST Framework, Streamlit

Infrastructure: Git, GitHub Actions, Docker, Jenkins, Kubernetes, AWS, Linux

Non-Technical Skills: Cross-functional collaboration

Achievements and Positions of Responsibility

Codex Router | 12 merged PRs duolahypercho/codex-router
- Authored 12 merged PRs; fixed quota-scope failover and subagent lifecycle handling, and added tool-schema regression tests.